

JSS 2 MATHEMATICS – FIRST TERM

TOPIC 8: ALGEBRAIC EXPRESSIONS I

1. Definition of Algebraic Expressions

An algebraic expression is a combination of numbers, letters (variables) and operation signs (e.g. $3x + 5$, $2a - b + 7$). Letters stand for unknown or varying quantities.

2. Terms, Coefficients and Like Terms

- Term: each part separated by + or – (e.g. in $3x + 5$, the terms are $3x$ and 5).
- Coefficient: the number multiplying the variable (in $3x$, coefficient is 3).
- Like terms: terms with the same variable(s) to the same power (e.g. $2x$ and $5x$; $3a^2$ and $-a^2$).

3. Expression of Algebraic Forms

Translate word statements into algebraic expressions. Example: “5 more than twice a number x ” $\rightarrow 2x + 5$.

4. Factorization of Simple Algebraic Expressions

Factorization means writing an expression as a product of factors. Common method: take out the highest common factor (HCF).

Example: $6x + 9 = 3(2x + 3)$

Example: $4a - 8b = 4(a - 2b)$

5. Summary

Algebraic expressions use letters and numbers. Identify terms and coefficients, collect like terms, and factorise by taking out the common factor.

End of Topic 8